# PepLab

Sequence, chemistry and structural information
for peptide selection.

Peptide teams face a practical decision: which candidates deserve the next experimental batch? Sequence, chemistry and structural information describe different aspects of a peptide. PepLab brings these sources into a shared selection workflow and makes their contributions inspectable.

## A measured example of source complementarity

**16% more top-eight hits** when contact features were added to sequence + chemistry, with the **same GP model, greedy selector and measurement budget**. Average discoveries rose from **2.815 to 3.265**.

Average top-eight hits found after 32 reveals

This is a saved retrospective development comparison: 160 HLA-A*02:01 nine-mers, 200 shared initializations, eight initial measurements and three batches of eight. A “hit” is one of the eight strongest measured binders within this finite cohort. Values were revealed from archived assays; no new assays were performed.

## Why combine sources?

**Sequence:** pretrained ESM2 representations capture sequence patterns. **Chemistry:** descriptors characterize molecular composition and physicochemical properties. **Contacts:** prepared structures describe peptide–receptor interaction patterns. **Direct energy:** Coulomb and Lennard-Jones terms provide a separate interaction proxy.

The study tested all 15 source subsets across two predictors and five selectors, plus random: 151 configurations and 30,200 episodes. Contact and energy features were evaluated separately. The observed leader used sequence + chemistry + contacts; adding every available source was not consistently best.

## The next experimental decision

Turn available peptide information into a ranked, traceable shortlist for the next experimental decision. Compare alternatives at the same budget and show which sources changed the recommendation. PepLab Bench is the public evidence and integration package behind that product direction.

# Selecting the next batch

The workflow, current implementation
and next technical steps.

## Workflow

**Input:** a defined candidate library, target and assay objective; available measurements; available sequence, chemistry and structural information; and an agreed experimental budget.

**Decision package:** a recommended candidate batch, alternative rankings, source-by-source contributions, uncertainty labels, a selection history and an exportable evidence file. These are the planned customer-facing outputs; the current public package supplies the benchmark, saved data, source interfaces and reproducible replay tools.

## Current implementation and planned extensions

| Available now | Product development |
|---|---|
| Public code, 160 candidate records, fixed feature blocks, all 151 method summaries, 280 matched source additions, figures and model integration contracts. | Cost-aware acquisition: decide which candidate and which evaluation layer to purchase next under explicit assay and compute budgets. |
| Transparent saved-results exploration and opt-in replay. Fresh-clone installation, 16 software/data checks and saved-results export verified. | Independent prospective evaluation, source-cost measurement and scalable candidate generation or retrieval before extending to very large design spaces. |

## Budgeted experimental planning

The development study supports the technical rationale for integrating complementary information. The next R&D; step is an engine that allocates a limited experimental and computational budget, learns from returned measurements and records why each action was selected. Its benefit should be tested against frozen cheap-only, random and fixed-pipeline policies on eligible independent evaluation data.

## Using another model

Bring a predictor or feature source, retain the same candidate pool and budget, and measure its incremental contribution. A provider can contribute sequence, structural or other representations without replacing the complete selection workflow. Covariance-dependent acquisitions require valid uncertainty outputs; mean-only models can support greedy selection.

## Evidence you can inspect

github.com/otcan/peplab-bench — code, peptide records, saved results, methods, figures and integration instructions.

Interpretation: the 16% comparison is an observed source addition on one previously exposed development cohort. The leading configuration was identified on that development matrix. Two hundred starts measure initialization variability, not independent biological replication. Structural features are template-informed; direct energies are proxies rather than binding free energies. The study does not validate cost-aware layer allocation or trillion-candidate performance.